Notation

Notation	*Meaning*	*Example*	*Other notation*
$\texttt{a}$	The character 'a'	a,b,c	a, b, c
$\texttt{foo}$	The string "foo"	$\texttt{bar}, \texttt{thing}$	*bar, foo*
w, x, y, z	Arbitrary string	$w \in \Sigma^*$	
ε	The empty string		λ
k	Numerical constant	$k, -3, 1, 42$	
n, m	Numerical variable	$n = 2m$	
L, S, A, B	Language, set	$L = \Sigma^*$	
$\emptyset$	The empty set		$\{\}$
Σ	An alphabet	$\Sigma = \{\texttt{a}, \texttt{b}\}$	
$\{w \mid \textit{foo}\}$	The set of all w such that *foo* holds	$\{w \mid w^{\textit{rev}} = w\}$	
$\texttt{a}^k$	k repetitions of the character a	$\texttt{a}^4 = \texttt{aaaa}$	a^k, a^4
w^k	k repetitions of the string w	$w^3 = www$	
A^k	The set of all strings with k arbitrary elements from the set A	$\{1, 2\}^2 = \{11, 12, 21, 22\}$	
A^*	Kleene star operation on A	$\{0, 1\}^* = \{\varepsilon, 0, 1, 00, \ldots\}$, $\texttt{a}^* = \{\varepsilon, \texttt{a}, \texttt{aa}, \ldots\}$, $(\texttt{foo})^* = \{\varepsilon, \texttt{foo}, \texttt{foofoo}, \ldots\}$	
A^+	Plus operator on the set A	$\{0, 1\}^+ = \{0, 1, 00, 01, 10, \ldots\}$	
$w^{\textit{rev}}$	Reverse of the string w	$(\texttt{foobar})^{\textit{rev}} = \texttt{raboof}$	w^R
$\overline{L}$	The complement language to L		$\neg(L), L^C$
2^A	The powerset of A	$2^{\{0,1\}} = \{\emptyset, \{0\}, \{1\}, \{0, 1\}\}$	
$\|A\|$	Number of elements in A (sometimes denoted *cardinality*)	$\|\{1, 2, 3\}\| = 3, \|\texttt{automation}\| = 7$	
$\|w\|_a$	The number of a's in the string w	$\|\texttt{abracadabra}\|_{\texttt{a}} = 5$	$n_a(w), \#_a(w)$
$\textit{prefix}(w)$	The set of strings x such that $w = xz$	$\textit{prefix}(\texttt{abc}) = \{\varepsilon, \texttt{a}, \texttt{ab}, \texttt{abc}\}$	
$\textit{pprefix}(w)$	The set of proper prefixes to w	$\textit{prefix}(\texttt{abc}) = \{\texttt{a}, \texttt{ab}\}$	
$\textit{prefix}(L)$	The set of strings w such that w is a prefix in some string in L	$\textit{prefix}(\Sigma^*) = \Sigma^*$	
$\textit{suffix}(w)$	The set of strings x such that $w = zx$	$\textit{suffix}(\texttt{abc}) = \{\varepsilon, \texttt{c}, \texttt{bc}, \texttt{abc}\}$	
$\textit{psuffix}(w)$	The set of proper suffixes to w	$\textit{psuffix}(\texttt{abc}) = \{\texttt{c}, \texttt{bc}\}$	
$\textit{suffix}(L)$	The set of strings w such that w is a suffix in some string in L	$\textit{suffix}(\Sigma^*) = \Sigma^*$	
$A \cup B$	The union of A and B	$\{1, 2, 3\} \cup \{2, 3, 4\} = \{1, 2, 3, 4\}$	$A + B$
$A \cap B$	The intersection of A and B	$\{1, 2, 3\} \cap \{2, 3, 4\} = \{2, 3\}$	
xy	The concatenation of the strings x and y	$x = \texttt{foo}, y = \texttt{bar}, xy = \texttt{foobar}$	
$A - B$	The set of all elements which are in A but not in B	$\{1, 2, 3\} - \{2, 3, 4\} = \{1\}$	$A \setminus B$
$A \times B$	The set of all combinations of an element in A concatenated with an element in B.	$\{\texttt{a}, \texttt{b}\} \times \{\texttt{c}, \texttt{d}\} = \{\texttt{ac}, \texttt{ad}, \texttt{bc}, \texttt{bd}\}$	
$\#$	The end of a string/stack		
$\mathbb{N}$	The set of natural numbers	$\mathbb{N} = \{0, 1, 2, \ldots\}$	
$\mathbb{Z}$	The set of integers	$\mathbb{Z} = \{\ldots, -2, -1, 0, 1, 2, \ldots\}$	
$\mathbb{R}$	The set of real numbers	$47,011 \in \mathbb{R}$	
$\therefore$	Ergo, hence, therefore		
$\square$	End of proof		$\blacksquare$